$$\mathcal{L}\{f(t)\} = \int_0^\infty f(t)\, e^{-st}\, dt$$

transformation

e^{at}

$$e^{at} \xrightarrow{\mathcal{L}} \frac{1}{s-a}$$

$$\int_0^\infty e^{-(s-a)t}\, dt = \left[\frac{e^{-(s-a)t}}{-(s-a)}\right]_0^\infty = \frac{1}{s-a}$$

$$\int_0^\infty \left[\frac{e^{-(s-a)t}}{-(s-a)}\right]' e^{-st} = \left[\frac{e^{-(s-a)t}}{-(s-a)} \cdot \frac{e^{-st}}{-s}\right]_0^\infty$$

$$\int_0^\infty \left[\frac{e^{-(s-a)t}}{-(s-a)} \cdot \frac{e^{-st}}{-s}\right]' = \left[\frac{e^{-(s-a)t}}{-(s-a)} \cdot \frac{e^{-st}}{-s} \cdot \frac{e^{-st}}{-s}\right]_0^\infty$$

$$1 + \frac{s}{s-a}$$

$$\mathcal{L}\{1 + e^{at}\}$$

$$\frac{1}{s-a}$$

$$\max = \frac{s}{s-a}$$

$$e^{-st} \cdot \frac{s}{s-a}$$

$$\max = \frac{s}{s-a}$$

$$\int_0^\infty \left[\frac{e^{-(s-a)t}}{-(s-a)} \cdot \frac{e^{-st}}{-s}\right]' = \left[\frac{e^{-(s-a)t}}{-(s-a)} \cdot \frac{e^{-st}}{-s} \cdot \frac{e^{-st}}{-s}\right]_0^\infty$$

t^n

$$\frac{n!\,\Gamma(\text{frac})}{s^{n+1}}$$

shift

$$\mathcal{L}\{e^{at} f(t)\}$$

$$\text{shift} \to F(s-a)$$

$$\text{insert } s \to s-a$$

proof $F(s-a) \to \mathcal{L}\{e^{at} f(t)\}$

$$t^{-\frac{1}{2}} \xrightarrow{\mathcal{L}(t^n)} \frac{\Gamma(\frac{1}{2})}{s^{\frac{1}{2}}} \to \frac{\sqrt{\pi}}{s^{\frac{1}{2}}}$$

$$\mathcal{L}\{u(t-a)\} \to \frac{e^{-as}}{s}$$

$$\mathcal{L}\{f(t)\}$$

$$e^{-as} F(s)$$

$$F(s) = \int_0^\infty f(t)\, e^{-st}\, dt$$

$$s = s - a(s)$$

$$s^2 F(s) - s^2 f(0) - s f'(0) \ldots$$

$$\text{no } s^2 \text{ or } s^3$$

sis replaced by $s-a$

while e^{-st} is present s no s is replaced

LHS → RHS

$$\int_0^\infty u(t-a) f(t-a)\, e^{-st}\, dt$$

$$\int_0^a 0 + \int_a^\infty f(t-a)\, e^{-st}\, dt$$

$$u = \text{new } t - a$$

$$\text{from} : \to u = 0$$

$$\text{to} \to u = +\infty$$

$$\int_0^\infty f(u)\, e^{-s(u+a)}\, du$$

$$e^{-sa} \int_0^\infty f(u)\, e^{-su}\, du$$

$$F(s)$$

$t^n f(t)$

$$(-1)^n \frac{d^n F(s)}{ds^n}$$

$$F(s) = \int_0^\infty e^{-st} f(t)\, dt$$

$$\frac{d F(s)}{ds} = \int_0^\infty \frac{d(e^{-st} f(t))}{ds}\, dt$$

$$= \int_0^\infty (-t) e^{-st} f(t)\, dt$$

$$= -\mathcal{L}\{t f(t)\}$$

$\frac{f(t)}{t}$

$$= \int_s^\infty F(s)\, ds$$

$$F(s) = \int_s^\infty \int_0^\infty f(t) e^{-st}\, dt\, ds$$

$$= \int_0^\infty \int_s^\infty f(t) e^{-st}\, ds\, dt$$

$$= \int_0^\infty \frac{f(t)}{-t} e^{-st}\, dt$$

$$= \mathcal{L}\left\{\frac{f(t)}{t}\right\}$$

$\int_0^t f(t)\, dt$

$$\left\{\frac{f(t)}{t}\right\} \to \mathcal{L}\{f(t)\}$$

$$= \frac{F(s)}{s}$$

step

$$u(t-a) = \begin{cases} 1 & t \geq a \\ 0 & t < a \end{cases}$$

$$\delta(t-a) = \begin{cases} \infty & t = a \\ 0 & t \neq a \end{cases}$$

$$f(t) = \begin{cases} f & a \leq t \leq b \\ 0 & \text{else} \end{cases}$$

$$f(t) = f[u(t-a) - u(t-b)]$$

$$= f[u(t-a) - u(t-b)]$$

$$= f[u(t-a) - u(t-b)]$$

$$= f[u(t-a) - u(t-b)]$$

$$= f[u(t-a) - u(t-b)]$$

$$= \frac{f}{s} e^{-as} - \frac{f}{s} e^{-bs}$$

$$= \frac{f}{s}(e^{-as} - e^{-bs})$$

$$8[1 - u(t-2)]$$

$$+ 6[u(t-3)]$$

$$8 + u(t-2) - 2$$

impulse

$$\mathcal{L}\{\delta(t-a)\} = e^{-as}$$

$$\text{sift} \to \int_0^\infty f(t) e^{-st} \delta(t-a)\, dt$$

$$= f(a) e^{-as}$$

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-as}} \int_0^a f(t) e^{-st}\, dt$$

$$= \frac{1}{1-e^{-as}} \left[\frac{-e^{-st}}{s} - \frac{e^{-st}}{s^2} + \frac{1}{s^2}\right]_0^a$$

periodic

$$\mathcal{L}\{f(t)\} = \frac{1}{1-e^{-as}} \int_0^a f(t) e^{-st}\, dt$$

$$f(t) = \begin{cases} f & 0 \leq t \leq a \\ -f & a \leq t \leq 2a \end{cases}$$

$$= \frac{1}{1-e^{-2as}} \int_0^a f e^{-st} - \int_a^{2a} f e^{-st}$$

$$= \frac{1}{1-e^{-2as}} \left[\frac{-f e^{-st}}{s}\right]_0^a + \left[\frac{f e^{-st}}{s}\right]_a^{2a}$$

$$f(t) = \begin{cases} f & 0 \leq t \leq a \\ -f & a \leq t \leq 2a \end{cases}$$

$$= \frac{1}{1-e^{-2as}} \int_0^a f e^{-st} - \int_a^{2a} f e^{-st}$$

$$= \frac{1}{1-e^{-2as}} \left[\frac{-f e^{-st}}{s}\right]_0^a + \left[\frac{f e^{-st}}{s}\right]_a^{2a}$$

$$= \frac{1}{1-e^{-2as}} \left[\frac{-f e^{-st}}{s} + \frac{f e^{-st}}{s}\right]_0^a$$

$$= \frac{1}{1-e^{-2as}} \int_0^a f e^{-st}$$

$$f(t) = \begin{cases} f & 0 \leq t \leq a \\ -f & a \leq t \leq 2a \end{cases}$$

$$f(t) = \frac{4}{\pi}\left(\sin t + \frac{\sin 3t}{3} + \frac{\sin 5t}{5} + \ldots\right)$$

$$f(t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\sin(2n-1)t}{2n-1}$$

$$f(t) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{\sin(2n-1)t}{2n-1}$$

$$\text{period} = 2\pi$$

$$\text{period} = x$$

partial

inverse

conv

$$F(s)\ G(s)$$

$$\text{conv } f(t) * g(t)$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2(s-1)}\right\} = \int_0^t f(u)\, g(t-u)\, du$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+1}\right\} = \int_0^t \cos u \cdot \frac{1}{s} \sin u(t-u)$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+1}\right\} \mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \int_0^t \cos u \cdot \sin u(t-u)$$

$$\cos u \cdot \frac{\sin u}{a} - \frac{\cos u}{a}$$

$$= \frac{1}{2a}$$

$$= \int_0^t \cos u \sin t + \cos u(t-2u)$$

$$= \frac{1}{2a} \left[\sin u \sin t + \frac{-\cos u(t-2u)}{-2a}\right]_0^t$$

$$= \frac{t \sin t}{2a}$$

$$= \frac{\cos t}{2a} - \frac{\cos(t-2t)}{2a}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-1)(s+1)}\right\} \mathcal{L}^{-1}\left\{\frac{1}{(s-1)^2}\right\} \cdot \left(\frac{1}{s-1}\right)$$

$$F(s) \quad G(s)$$

$$= \int_0^t f(u)\, g(t-u)\, du$$

$$\frac{A}{(s+1)} + \frac{B}{(s+1)^2} + \frac{C}{(s-1)}$$

$$A(s+1)(s-1) + B(s-1) + C(s+1)^2$$

$$s^2: \quad A + C = 0$$

$$s^1: \quad 2A + B + 2C = 0$$

$$s^0: \quad -A - B + C = 1$$

$$2A + B = -1$$

$$4A = -1$$

$$A = -\frac{1}{4}$$

$$B = -\frac{1}{2}$$

$$C = \frac{1}{4}$$

$$\frac{1}{(s+1)^2 - 1^2} \Rightarrow \sinh(t) e^{-t}$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+1} \cdot \frac{e^{-\theta s}}{s^2+1}\right\} = u(t-\theta) + (t-\theta) \cos(t-\theta)$$

$$\text{RHS}: \text{const}$$

$$\frac{(s+1)-1}{(s+1)^2+1}$$

$$e^{-t} \cos t - \sin t\, e^{-t}$$

$$\mathcal{L}[f(s) * G(s)] = \int_0^t f(u)\, g(t-u)\, du$$

$$\int_0^t u e^{t-2u} \to e^t \int_0^t u e^{-2u}$$

$$e^t \left[u \frac{e^{-2u}}{-2} - \frac{e^{-2u}}{4}\right]_0^t$$

$$e^t \left[\frac{t e^{-2t}}{-2} - \frac{e^{-2t}}{4} + \frac{1}{4}\right]$$

$$\frac{t e^{-t}}{-2} - \frac{e^{-t}}{4} + \frac{e^t}{4}$$

$$e^{2t} * \sin t$$

$$\int_0^t \sin u\, e^{2(t-u)}$$

$$e^{2t} \int_0^t e^{-2u} \sin u$$

$$\int_0^t t e^{-2u}\, du \Rightarrow \left[\frac{t e^{-2u}}{-1}\right]_0^t$$

$$-\left[t e^{-2t} - t\right]$$

cramer

$$a F(s) + b G(s) = A$$

$$c F(s) + d G(s) = B$$

$$F(s) = \frac{\begin{vmatrix} A & b \\ B & d \end{vmatrix}}{D}$$

$$D = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$F(s) = \frac{\begin{vmatrix} A & b \\ B & d \end{vmatrix}}{D}$$

$$G(s) = \checkmark$$

BDD Must given

application

diff → simul

$$\frac{d^2 y}{dx^2} - \frac{dy}{dx} - 2y = 200 \cos 3x - 100 \sin 3x$$

$$\mathcal{L}\{y\} = F(s)$$

$$\mathcal{L}\{\ddot{y} + \dot{y} - 2y\} = \mathcal{L}\{200\cos 3x - 100\sin 3x\}$$

$$\mathcal{L}\{\ddot{y}\} + \mathcal{L}\{\dot{y}\} - 2\mathcal{L}\{y\} = \mathcal{L}\{200\cos 3x\} - \mathcal{L}\{100\sin 3x\}$$

$$\mathcal{L}\{\ddot{y}\} = s^2 F(s) - s y(0) - y'(0)$$

$$\mathcal{L}\{\dot{y}\} = s F(s) - y(0)$$

$$\mathcal{L}\{y\} = F(s)$$

$$s^2 F(s) - s y(0) - y'(0) - [s F(s) - y(0)] - 2F(s) = \frac{200 s}{s^2+9} - \frac{300}{s^2+9}$$

$$F(s)[s^2 - s - 2] - b = \frac{200(s-1)}{s^2+9}$$

$$\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{200(s-1) + b(s^2+9)}{(s^2+9)(s^2-s-2)}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{200s - 200 + bs^2 + 9b}{(s^2+9)(s^2-s-2)}\right\}$$

$$s^3: \quad 0 + b = 0$$

$$s^2: \quad 200 + 9b = 0$$

$$s^1: \quad 200 + 2b = 0$$

$$s^0: \quad 200 - 18b = 0$$

$$b = -\frac{200}{9}$$

$$\text{ex(1)} \quad \text{let } F(s) = \mathcal{L}\{y\}$$

$$\mathcal{L}\{2y'' - 3y' + 2x\} = \mathcal{L}\{1 - e^{2x}\}$$

$$\mathcal{L}\{2y''\} - 3\mathcal{L}\{y'\} + 2\mathcal{L}\{x\} = \mathcal{L}\{1\} - \mathcal{L}\{e^{2x}\}; \text{ linear prop}$$

$$2s^2 F(s) - 2s y(0) - y'(0) - 3[s F(s) - y(0)]$$

$$= \frac{2}{s} - \frac{1}{s-2}$$

$$F(s)[2s^2 - 3s] = \frac{2(s-2) - s}{s(s-2)} - \frac{2}{s^2}$$

$$= \frac{-2s^2 - 2s^2 + 2s}{s^2(s-2)} \cdot \frac{s(s-3)}{s(s-3)}$$

$$= \frac{-4s^2 + 2}{s^3(s-2)(s-3)}$$

$$= \frac{2(s^2-2)}{s^3(s-2)(s-3)}$$

$$\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{2}{s^3(s-3)}\right\}$$

$$\mathcal{L}^{-1}\left\{\int_0^t f(u)\, du\right\} = \frac{F(s)}{s}$$

$$\mathcal{L}^{-1}\left\{\frac{2}{s-3}\right\} = 2e^{3t}$$

$$\mathcal{L}^{-1}\left\{\frac{2}{s^2(s-3)}\right\} = \int_0^t 2e^{3u}\, du$$

$$= 2\left[\frac{e^{3u}}{3}\right]_0^t$$

$$= \frac{2}{3}(e^{3u} - 1)$$

$$\mathcal{L}^{-1}\left[\frac{\frac{2}{s^2(s-3)}}{s}\right] = \int_0^t \frac{2}{3}(e^{3u} - 1)$$

$$= \frac{2}{3}\left[\frac{e^{3t}}{3} - t\right]_0^t$$

$$= \frac{2}{3}\left[\frac{e^{3t}}{3} - \frac{1}{3} - t\right]$$

$$\mathcal{L}^{-1}\left\{\frac{\frac{2}{s^2(s-3)}}{s}\right\} = \frac{2}{3}\left[\int \cdots\right.$$

$$= \frac{2}{3}\left[\frac{e^{3t}}{9} - \frac{t}{3} - \frac{t^2}{2}\right]_0^t$$

$$\frac{2}{3}\left[\frac{e^{3t}}{9} - \frac{1}{9} - \frac{t}{3} - \frac{t^2}{2}\right]$$

$$\frac{1}{s^{1/2}}$$

$$m+1 = \frac{1}{2}$$

$$n = -\frac{1}{2}$$

$$\frac{\Gamma(1/2)}{s^{-\frac{1}{2}+1}} = \frac{1}{\sqrt{\pi}} t^{-\frac{1}{2}}$$

$$\text{g) 3)} \quad \text{let } \mathcal{L}\{u(t)\} = F(s)$$

$$\mathcal{L}\{y(t)\} = G(s)$$

$$\text{convolution } \mathcal{L}$$

$$\mathcal{L}\{\mathcal{L}\{x\} * \mathcal{L}\{x\}\} + \mathcal{L}(G(s-y))$$

bendy beam

$$EI \frac{d^4 y(x)}{dx^4} = \infty\, \delta(x - \tfrac{L}{4})$$

$$= \infty\, \delta(x)$$

$$\text{unit} \checkmark$$

cct

$$\begin{cases} \text{ibik} \\ L \frac{di}{dt} \\ \frac{1}{C} \int i\, dt \end{cases} \begin{cases} \text{charge} \\ i = \frac{dq}{dt} \\ \text{input } j; \end{cases}$$

spring/mass

$$F(s) = m\ddot{x} + c\dot{x} + kx$$

$$\frac{d}{dt}\{u(t-a)\} = \delta(t-a)$$

water/ctm

nuclear physics